

Curriculum Vitae of Katarzyna Wręczycka, PhD

 [katwre.github.io](https://github.com/katwre)  scholar.google.com/citations?user=katarzynawreczycka  github.com/katwre

Profile

Computational biologist and machine-learning researcher with a PhD in Computational Molecular Biology and 9+ years of experience across academia and industry.

Work Experience

Computational Biologist & Machine Learning Scientist

Freelance | Freiberufler

02.2023 – present
Berlin, Germany 
Zurich, Switzerland 

Engaged in B2B contracts with pharmaceutical companies, start-ups and R&D consultancy companies. Key highlights include:

- **Anomaly detection for water safety monitoring** 01–06.2026
Developed an LSTM-based variational autoencoder in PyTorch to detect and forecast anomalous patterns of chemicals in high-frequency laser particle-counter measurements, and contributed to a Kafka-based streaming pipeline.
- **Menstrual blood biomarker discovery** 09–12.2025
Led a pilot project on classifying endometriosis, blood profiling, oncology biomarker discovery and immune cell dynamics using targeted RNA-seq and scRNA-seq from menstrual effluent via using generative modeling and transfer learning (scanpy, scvi-tools, scArches) and evaluated downstream classifiers (XGBoost) for endometriosis-associated signatures on AWS (EC2, S3).
- **Oncology biomarker discovery in healthcare** 05.2023–01.2025
 - **Machine learning for drug development:** Applied positive unlabeled learning with random forests and SVM (with pulearn) and autoencoder (in PyTorch) to prioritize transcription factors for future clinical trials, integrated clinical-trial insights from GenomOncology database and public databases such as DepMap, TCGA, TEMPUS.
 - **Translational oncology signature discovery:** Collaborated closely with data scientists and experimental scientists in R&D and Clinical Development teams to identify tumor-associated antigens and gene signatures for patients with limited treatment options. Leveraged RNA-seq, WGS, external clinical trial databases, and performed survival analysis to support therapeutic decision-making.
- **Multimodal omics analysis in Alzheimer's disease** 03–05.2023
Investigated glia-to-neuron reprogramming by integrating gene expression and epigenomic data, including RNA-seq, ATAC-seq, ChIP-seq and DNA methylation data in reproducible Nextflow/nf-core workflows. Applied the hybrid CNN–Transformer Enformer model (Avsec et al., Nature Methods, 2021) to characterize regulatory enhancers and their potential target genes, and presented findings to wet-lab scientists to inform downstream experimental validation.

Computational Discovery Scientist

Polaroid Therapeutics

06.2026 – present
Zurich, Switzerland 

- **Crohn's disease biomarker discovery:** Developing computational and machine-learning approaches for biomarker discovery (incl. mixed-effect regression, XGBoost) in Crohn's disease, integrating large-scale biobank data from the IBD Plexus of the Crohn's & Colitis Foundation, including Olink (proteomics) from blood samples.
- Developed AWS infrastructure for sensitive biomedical data analysis.

Bioinformatician | Research Assistant II

ETH Zürich & Functional Genomics Center Zürich (FGCZ) of the University of Zurich (UZH)
Employed by the Prof. von Meyenn Lab (40% position), Prof. Bar-Nur Lab (40%) and FGCZ/UZH (20%)

04.2025 – 06.2025
Zurich, Switzerland 

- **Integrative omics analyses across multiple projects:** Performed gene-regulatory analyses for muscle aging with natural compounds (RNA-seq), germline chimerism (snRNA-seq), yo-yo effect in adipocytes (CUT&TAG), and assembly of Y chromosome of *Mus minutoides* (WGS). Worked closely with wet-lab teams to interpret results and translate analyses into downstream experimental follow-up.

- **Reproducible NGS workflows on HPC:** Maintained and operated end-to-end Nextflow pipelines (RNA-seq, CUT&TAG) via Nextflow Tower on HPC (SLURM) at ETH Zürich, using Docker/Singularity and Spack-managed environments.
- **ShinyProxy for snRNA-seq:** Deployed interactive R Shiny apps via ShinyProxy for snRNA-seq datasets (shiny-public.fgc.z.uzh.ch ; snRNA-seq: MHUO, Mouse epiAT memory, Human AT memory).

PhD Student in Computational Molecular Biology

Berlin Institute for Medical Systems Biology (BIMSB) at the Max Delbrück Center for Molecular Medicine (MDC) & Humboldt University of Berlin.

10.2015 – 11.2021
Berlin, Germany 

- Led and contributed to research projects applying statistical and ML methods, resulting in 4 **publications with 420+ citations**.
- **Genomic artifact detection:** Investigated high-density transcription factor binding sites (HOT regions) and identified spurious ChIP-seq peaks linked to RNA structures called DNA-RNA hybrids (R-loops), and hypo-methylated regions using elastic net regression and PCA. Integrated data from publicly available resources, including ENCODE and Roadmap Epigenomics, as well as experimental datasets from ChIP-seq, ATAC-seq, Bisulfite-seq, and DRIP-seq/RDIP-seq studies  [2].
- **DNA methylation biomarkers for acute coronary syndrome discovery:** Studied Bisulfite-seq data from patient-derived heart tissue and plasma, uncovering gene and transcription factor associations. My contribution included preprocessing raw Bisulfite-seq data, calling differentially methylated regions, associating them with genes, and transcription factors in cardiovascular diseases [1].
- **Snakemake pipeline for Bisulfite-seq:** A scalable pipeline using TrimGalore!, BWA-meth/Bismark, samblaster, MethylKit/methylDackel, and genomation R packages, published as part of the PiGX: Pipelines in Genomics project  [3].
- First co-author of a peer-reviewed **review on statistical methods for analyzing Bisulfite-seq data**  [4].
- **Developed the R package motifActivity** for transcription factor network reconstruction from sequencing data using rank-based estimation linear models .
- **Multi-omics analysis of high-risk pediatric neuroblastoma:** Identified DNA methylation biomarkers and transcription factor networks specific to high-risk cases from solid tumors and cell-free DNA in urine. Applied machine learning methods (incl. elastic net, ridge regression, XGBoost) and integrated patient-derived samples from Bisulfite-seq, RNA-seq, and ChIP-seq datasets in collaboration with Dr. med. Johannes H. Schulte from Charité Hospital in Berlin.

Scholarships

Received a €7,000 scholarship from the Institute of Computer Science, Polish Academy of Sciences (IPI PAN), co-funded by the European Union funds, for predoctoral research at the MDC-BIMSB. Also co-developed genomation, a Bioconductor tool for genomic data visualization and annotation .

03 – 09.2015
Berlin, Germany 

Education

Ph.D. in Computational Molecular Biology

Humboldt University of Berlin

Advisors: Dr. Altuna Akalin and Prof. Dr. Uwe Ohler

Thesis: "Integrative analysis on the effect of DNA methylation on gene regulation in tumor and healthy tissues" (Link to PDF).

10.2015 – 05.2021
Berlin, Germany 

M.Sc. in Bioinformatics

University of Warsaw, Department of Mathematics, Informatics and Mechanics

Advisor: Prof. Dr hab. Anna Gambin

Thesis: "Genome-wide analysis of noncoding copy-number variations in human genome".

10.2012 – 06.2014
Warsaw, Poland 

B.Sc. in Bioinformatics

University of Warsaw, Department of Mathematics, Informatics and Mechanics

Advisor: Dr hab. Bartosz Wilczynski

Thesis: "Potential regulatory region discovery method implementation" .

10.2010 – 06.2012
Warsaw, Poland 

- [1] R. R. C. Cuadrat, A. Kratzer, H. G. Arnal, R. A. C., K. Wreczycka, A. Blume, V. Ebenal, T. Mauno, B. Osberg, M. Moobed, J. Hartung, C. Seppelt, D. Meteva, A. Haghikia, D. Leistner, U. Landmesser and A. Akalin, "Cardiovascular disease biomarkers derived from circulating cell-free dna methylation," *NAR Genomics and Bioinformatics*, 2023. doi: 10.1093/nargab/1qad061.
- [2] K. Wreczycka, V. Franke, B. Uyar, R. Wurmus, S. Bulut, B. Tursun and A. Akalin, "HOT or not: examining the basis of high-occupancy target regions," *Nucleic Acids Research*, vol. 47, no. 11, pp. 5735–5745, May 2019. doi: 10.1093/nar/gkz460.
- [3] R. Wurmus, B. Uyar, B. Osberg, V. Franke, A. Gosdschan, K. Wreczycka, J. Ronen and A. Akalin, "PiGx: reproducible genomics analysis pipelines with GNU Guix," *GigaScience*, vol. 7, no. 12, Oct. 2018, giy123. doi: 10.1093/gigascience/gy123.
- [4] K. Wreczycka, A. Gosdschan, D. Yusuf, B. Grüning, Y. Assenov and A. Akalin, "Strategies for analyzing bisulfite sequencing data," *Journal of Biotechnology*, vol. 261, pp. 105–115, 2017, *Bioinformatics Solutions for Big Data Analysis in Life Sciences presented by the German Network for Bioinformatics Infrastructure*. doi: 10.1016/j.jbiotec.2017.08.007.

Conferences & workshops

Talks & Conference Presentations

- **Speaker**, "Assessing prognostic and diagnostic features of neuroblastoma methylome", DTKT – Young Academics Meeting, 6 Sep. 2019, Berlin, Germany
- **Poster**, "Assessing prognostic and diagnostic features of high-dimensional (epi)genomics data for neuroblastoma", RECOMB, 5–8 May 2018, Washington, D.C., USA
- **Speaker**, "HOT or not: Examining the basis of high-occupancy target regions", CRG-BIMSB MDC PhD Retreat, 15–18 May 2017, Girona, Spain
- **Poster presentation**, "HOT or not: redefining the origin of high-occupancy target regions", EMBL Conference "From Functional Genomics to Systems Biology", 12–15 Nov. 2016, Heidelberg, Germany
- **Speaker**, "Genomation: an R package to summarize, annotate and visualize genomic intervals", Bioinformatics Social Meetings Berlin, 16 Sep. 2016, Berlin, Germany
- **Poster presentation**, "Genomation: an R package to summarize, annotate and visualize genomic intervals", Max-Planck-Gesellschaft Otto Warburg Summer School, 6–11 Sep. 2015, Berlin, Germany

Workshops & Training (selected)

- Parallelisation with OpenMP, Swiss Institute of Bioinformatics workshop, 11 May. 2025, Switzerland (📄, 🔄 GPU computing)
- Ensuring More Accurate, Generalisable, and Interpretable Machine Learning Models for Bioinformatics, Swiss Institute of Bioinformatics workshop, 17 Nov. 2025, Switzerland (📄, 🔄 XG-Boost Bayesian optimization)
- Machine Learning Zoomcamp, Data Talks Oct 2025 - Jan 2026 (📄)
 - Gene type prediction from DNA sequence: Designed and trained a transformer encoder for genomic sequence classification; implemented ONNX Runtime for inference and deployed a FastAPI and BentoML service on AWS Kubernetes 🔄.
 - Immune cell classifier: Developed and trained a CNN-based image classifier for immune cell detection in H&E-stained histopathological images using Xception with an MLP head; dockerized; deployed on AWS Lambda for scalable inference 🔄.
 - Diffusion models for immune cell images: Developed a UNet-based diffusion model with linear noise scheduling for H&E-stained immune cell image inpainting, with an interactive Streamlit frontend for model exploration, deployed on AWS EC2 🔄.
- Using Large Language Models for Biodata Exploration: From Theory to Practice, Swiss Institute of Bioinformatics workshop, 30 Sept. 2025, Zurich, Switzerland (🔄 LLM-based AI assistants)
- PyData Berlin Conference and Workshops, 1–3 Sep. 2025, Berlin, Germany (🔄 State-Space Models with PyMC)
- Federated Learning in Bioinformatics, Swiss Institute of Bioinformatics workshop, 29 Aug. 2025, Lugano, Switzerland (📄, 🔄 VAE to mitigate batch effects in scRNA-seq using federated learning)
- EMBL-EBI Cancer Genomics course, 20–23 June 2017, Hinxton, United Kingdom